Part X - Rheumatology and Immunology
Chapter 10B: Rheumatoid Arthritis, Spondyloarthritis, Psoriatic Arthritis, and Reactive Arthritis

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Scope of this chapter

This chapter follows the 10A overview by focusing on disease-specific inflammatory arthritis patterns. Rheumatoid arthritis is used as the prototype of symmetric small-joint synovitis. The spondyloarthritis family is used as the prototype of seronegative disease with axial inflammation, enthesitis, dactylitis, psoriasis, uveitis, inflammatory bowel disease, or a post-infectious trigger. The uploaded rheumatology source pages emphasize RA features such as MCP/PIP/wrist involvement, DIP sparing, morning stiffness, hand deformities, cervical instability, nodules, RF/anti-CCP interpretation, and extra-articular disease; this chapter expands those concepts into diagnosis and treatment algorithms.

10B.1 Core Pattern Recognition: RA vs Spondyloarthritis

Inflammatory arthritis becomes easier when it is sorted by time course, joint distribution, symmetry, axial involvement, serology, and extra-articular disease. Rheumatoid arthritis usually announces itself as chronic, symmetric, additive inflammatory polyarthritis of small joints, especially wrists, metacarpophalangeal joints, proximal interphalangeal joints, and metatarsophalangeal joints. The distal interphalangeal joints are usually spared. Spondyloarthritis often behaves differently: it may be asymmetric, lower-extremity predominant, associated with inflammatory back pain, or dominated by enthesitis and dactylitis rather than classic synovial swelling.

A useful first step is to ask whether the patient has true arthritis rather than arthralgia. True arthritis implies objective synovitis: swelling, warmth, restricted range of motion from effusion or synovial thickening, and pain at rest or with gentle passive motion. Arthralgia without swelling may still precede autoimmune arthritis, but it also points toward viral illness, endocrine disease, fibromyalgia, medication effects, periarticular disease, and early connective tissue disease.

Pattern	Most likely direction	High-yield clues	Common next tests
Symmetric small-joint inflammatory polyarthritis	Rheumatoid arthritis; also SLE, viral arthritis, early PsA	MCP/PIP/wrist swelling, morning stiffness >30-60 min, nodules, fatigue, erosions; DIP sparing favors RA	RF, anti-CCP, ESR/CRP, CBC/CMP, hand/foot imaging; screen before DMARDs
Asymmetric lower-extremity oligoarthritis	Reactive arthritis, PsA, peripheral SpA, crystal or septic arthritis	Knee/ankle/toe disease, enthesitis, dactylitis, recent diarrhea or urethritis/cervicitis, conjunctivitis, psoriasis	Arthrocentesis if effusion; Chlamydia/gonorrhea NAAT if exposure; ESR/CRP; targeted imaging
Inflammatory back pain	Axial spondyloarthritis / ankylosing spondylitis	Onset before 45, insidious, better with exercise, worse with rest, night pain, alternating buttock pain, uveitis	Pelvic radiograph; MRI SI joints if early/non-radiographic disease suspected; CRP; selective HLA-B27
DIP arthritis with nail disease	Psoriatic arthritis until proven otherwise	Nail pitting, onycholysis, hidden psoriasis, dactylitis, enthesitis	Skin/nail exam, RF/anti-CCP for RA mimic, radiographs for erosive/proliferative change
Explosive monoarthritis	Septic arthritis or crystal arthritis first	Fever, immunosuppression, prosthetic joint, severe passive-motion pain, overlying cellulitis	Arthrocentesis: cell count, Gram stain/culture, crystals; blood cultures if febrile

10B.2 Rheumatoid Arthritis: Disease Concept

Rheumatoid arthritis is a chronic systemic autoimmune disease centered on synovial inflammation. Synovium is not a passive lining in RA; it becomes a proliferative inflammatory tissue that invades cartilage and bone. Activated macrophages, T cells, B cells, fibroblast-like synoviocytes, osteoclast signaling, cytokines, chemokines, and autoantibodies create a self-amplifying pannus that produces erosions, tendon injury, deformity, pain, and systemic inflammation.

The key clinical consequence is that structural damage may begin early, so diagnosis and disease-modifying treatment should not wait for classic deformities. The typical age of onset is adulthood, often between the third and sixth decades, with female predominance. Smoking is an important environmental risk factor, particularly for anti-citrullinated protein antibody positive disease. Genetic susceptibility, especially certain HLA-DRB1 shared epitope alleles, increases risk, but RA remains a clinical diagnosis supported by serologies and imaging rather than a genetic diagnosis.

High-yield source anchor

The uploaded rheumatology source describes RA as a chronic inflammatory autoimmune synovial disease; emphasizes that inflammation can damage cartilage and bone; highlights symmetric polyarthritis involving MCP, PIP, wrist, knee, ankle, elbow, hip, and shoulder joints; and notes that DIP involvement is not typical RA. The source also stresses early DMARD treatment because disabling joint damage can occur early.

Pathogenesis and tissue injury

RA may begin before obvious arthritis. Mucosal immune activation in the lung, oral cavity, or gut can generate citrullinated proteins and autoantibodies. Anti-CCP antibodies may predate arthritis by years. Once synovitis develops, the inflamed synovium produces tumor necrosis factor, interleukin-6, interleukin-1, matrix metalloproteinases, and receptor activator of nuclear factor kappa-B ligand. These mediators recruit immune cells, stimulate osteoclasts, and degrade cartilage.

The term pannus refers to invasive hypertrophic synovium that erodes adjacent cartilage and bone. Tendon sheaths can become inflamed, causing tenosynovitis and tendon rupture. Ligamentous laxity and joint subluxation produce deformities such as ulnar deviation, swan-neck deformity, boutonniere deformity, and thumb deformities. These are late findings and should not be required before diagnosis.

Clinical features of RA

RA classically presents with gradual onset pain, swelling, warmth, and stiffness in multiple joints. Morning stiffness usually lasts more than 30 to 60 minutes and improves with activity. Symptoms are often additive and persistent rather than migratory. Fatigue, low-grade fever, anorexia, and weight loss reflect systemic inflammation. Joint swelling is usually soft or boggy because it arises from synovitis and effusion rather than bony osteophytes.

The highest-yield joint distribution is bilateral MCP, PIP, wrist, and MTP involvement. DIP disease is not impossible, but prominent DIP arthritis should trigger reconsideration of psoriatic arthritis, erosive osteoarthritis, gout, or hemochromatosis arthropathy. RA can involve knees, ankles, elbows, shoulders, hips, temporomandibular joints, cricoarytenoid joints, and the cervical spine. Atlantoaxial instability can injure the spinal cord, especially during intubation or neck manipulation.

RA finding	Mechanism or interpretation	Clinical consequence
MCP/PIP/wrist synovitis	Small-joint synovial inflammation and tenosynovitis	Classic pattern; supports RA when symmetric and persistent
MTP tenderness or squeeze-test pain	Forefoot synovitis is common and may precede hand deformity	Ask about walking pain; examine feet even when hands dominate symptoms
Ulnar deviation at MCPs	Chronic synovitis, ligament laxity, tendon imbalance, subluxation	Late uncontrolled disease; functional impairment
Boutonniere deformity	PIP flexion with DIP hyperextension from extensor mechanism injury	Suggests chronic structural damage
Swan-neck deformity	PIP hyperextension with DIP flexion from ligament/tendon imbalance	Suggests chronic structural damage; can impair grip
Rheumatoid nodules	Granulomatous nodules usually in seropositive disease, often on extensor surfaces	Associated with more severe extra-articular phenotype; may also occur in lung/pleura/pericardium
C1-C2 subluxation	Cervical synovitis and ligament destruction	Screen high-risk patients before surgery/intubation; neurologic signs are urgent
Felty syndrome	RA with neutropenia and splenomegaly	Infection risk; usually long-standing seropositive RA
Pleural effusion with very low glucose	Rheumatoid pleuritis	Distinguish from infection and malignancy; evaluate respiratory symptoms

Extra-articular RA

RA is systemic. Extra-articular disease is more likely in patients with high-titer RF or anti-CCP antibodies, smoking history, nodules, severe longstanding inflammation, and erosive disease. Eye disease includes dry eye, episcleritis, and scleritis; scleritis is painful and can threaten vision. Lung disease includes pleural effusion, interstitial lung disease, airway disease, pulmonary nodules, and drug-related pneumonitis. Cardiac involvement can include pericarditis, accelerated atherosclerosis, conduction disease, and valvular disease.

Anemia of chronic inflammation, thrombocytosis, hypoalbuminemia, and elevated ESR/CRP often track active inflammation. Neutropenia in a patient with long-standing RA should raise concern for Felty syndrome, large granular lymphocyte leukemia, medication toxicity, infection, or hypersplenism. Because RA therapies also affect immune defense, every systemic symptom must be sorted into active disease, drug toxicity, infection, malignancy, and unrelated pathology.

Organ system	RA manifestations	High-yield implication
Constitutional	Fatigue, malaise, low-grade fever, anorexia, weight loss	Can reflect inflammation, infection, malignancy, or drug toxicity; fever should not be automatically called flare
Skin/subcutaneous	Rheumatoid nodules, vasculitic ulcers, palpable purpura, digital ischemia	Nodules favor seropositive disease; ulcers/ischemia suggest vasculitis or embolic/antiphospholipid disease
Lung/pleura	Pleural effusions, interstitial lung disease, airway disease, nodules; methotrexate pneumonitis mimic	New dyspnea requires infection/drug/ILD evaluation; rheumatoid pleural fluid can have very low glucose
Eye	Keratoconjunctivitis sicca, episcleritis, scleritis, peripheral ulcerative keratitis	Scleritis and ulcerative keratitis are urgent because they can threaten vision
Nervous system	Carpal tunnel, mononeuritis multiplex, cervical myelopathy from instability	Focal neurologic deficits are red flags; cervical disease matters before intubation
Hematologic	Anemia of inflammation, thrombocytosis, neutropenia in Felty syndrome, medication cytopenias	CBC monitoring is part of disease assessment and drug safety

Organ system	RA manifestations	High-yield implication
Cardiovascular	Pericarditis, accelerated atherosclerosis, conduction abnormalities, valvular disease	Control inflammation and manage traditional cardiovascular risk aggressively

10B.3 Diagnosis of Rheumatoid Arthritis

RA is diagnosed by integrating clinical synovitis, distribution, symptom duration, serology, inflammatory markers, and imaging. No single test is definitive. Anti-CCP is highly specific and predicts erosive disease, but seronegative RA exists. RF is helpful when positive in the correct clinical context, especially at high titer, but it is less specific and may appear in chronic infections, other autoimmune diseases, older adults, and healthy individuals.

The 2010 ACR/EULAR classification framework uses joint involvement, serology, acute-phase reactants, and symptom duration to classify RA in patients with definite clinical synovitis not better explained by another disease. Classification criteria are not a substitute for clinical judgment. A patient with classic persistent symmetric MCP/PIP synovitis may be treated while workup is ongoing. Conversely, a positive RF in a patient with diffuse pain but no synovitis should not be labeled RA.

Component	What supports RA	Limitations and pitfalls
History	Morning stiffness >30-60 min, persistent symptoms >6 weeks, additive small-joint involvement, functional impairment	Stiffness can occur in many conditions; pain alone is not synovitis
Exam	Boggy swelling of MCP/PIP/wrist/MTP joints; symmetric synovitis; reduced fist closure; tendon sheath involvement	Obesity, edema, OA nodes, and periarticular pain can obscure exam; ultrasound can help
RF	High-titer positivity supports RA and more severe extra-articular phenotype	Not specific; can be positive in infections, other autoimmune disease, older adults, and healthy people
Anti-CCP/ACPA	High specificity for RA; predicts erosive disease and future RA in at-risk patients	Not present in all RA; rarely positive in other diseases
ESR/CRP	Elevated markers support inflammatory disease and help monitor activity	Normal values do not exclude RA; infection/malignancy also elevate markers
Radiographs	Marginal erosions, periarticular osteopenia, symmetric joint-space narrowing, subluxations	May be normal early; ultrasound/MRI detect earlier synovitis/erosions
Synovial fluid	Inflammatory fluid; useful when infection/crystal disease is possible	Not diagnostic for RA; always culture if septic arthritis is possible

Board-style trap

Positive RF plus joint pain is not RA unless there is inflammatory arthritis. Conversely, negative RF does not exclude RA. Use the joint pattern and synovitis exam first; use serology as supporting evidence.

RA differential diagnosis

Condition	How it mimics RA	Clues against RA
Osteoarthritis	Hand pain and stiffness, Heberden/Bouchard nodes, reduced function	Stiffness usually brief; bony enlargement; DIP and first CMC involvement; pain worse with use
Psoriatic arthritis	Polyarthritis can be symmetric and small-joint predominant	Psoriasis/nail pitting, dactylitis, enthesitis, DIP involvement, asymmetric pattern, axial disease
SLE arthritis	Symmetric inflammatory polyarthritis of hands/wrists	Nonerosive, migratory, systemic lupus features, ANA/dsDNA/Sm/complement abnormalities
Viral arthritis	Acute symmetric small-joint polyarthritis	Abrupt onset, exposure/prodrome, rash, hepatitis/parvovirus/chikungunya features; often self-limited
Crystal arthritis	Recurrent inflammatory attacks; polyarticular gout/CPPD may mimic RA	Abrupt flares, crystals on synovial fluid, tophi/chondrocalcinosis; asymmetric episodes
Septic arthritis	Hot swollen joint in immunosuppressed RA patient	Usually severe monoarthritis; fever may be absent; culture and urgent drainage/antibiotics
Fibromyalgia	Widespread pain and fatigue	No objective synovitis, normal inflammatory markers, tenderness is soft-tissue predominant

10B.4 Treatment of Rheumatoid Arthritis

The treatment objective in RA is remission or low disease activity, not merely tolerable pain. Treat-to-target care means measuring disease activity, starting a disease-modifying antirheumatic drug promptly, escalating therapy if the target is not reached, minimizing long-term glucocorticoids, and monitoring toxicity. Early methotrexate-based therapy remains a common first-line anchor for moderate-to-high activity RA when not contraindicated.

NSAIDs and corticosteroids are symptom relievers, not adequate stand-alone disease modifiers. Short glucocorticoid bridging can be helpful when starting or escalating DMARDs, but chronic steroids increase infection, osteoporosis, diabetes, hypertension, cataracts, avascular necrosis, adrenal suppression, and cardiovascular risk. A patient who repeatedly needs prednisone bursts is not controlled; that should trigger reassessment of diagnosis, adherence, dose optimization, infection, and DMARD escalation.

Medication class	Examples	Role in RA	Major safety/monitoring points
Conventional synthetic DMARDs	Methotrexate, hydroxychloroquine, sulfasalazine, leflunomide	Foundation therapies; methotrexate is common first-line anchor for active RA	CBC/CMP monitoring for MTX/leflunomide/sulfasalazine; folic acid with MTX; retinal screening for hydroxychloroquine; pregnancy/liver/lung considerations
Biologic DMARDs: TNF inhibitors	Etanercept, adalimumab, infliximab, golimumab, certolizumab	Escalation when csDMARD inadequate; also used in SpA/PsA	Screen TB/hepatitis; infection risk; caution with demyelinating disease and some heart failure contexts
Biologic DMARDs: non-TNF	Abatacept, rituximab, tocilizumab/sarilumab	Alternative mechanisms for inadequate response or specific comorbidity patterns	Infection risk; drug-specific monitoring such as lipids/LFTs/neutrophils for IL-6 blockade; hepatitis B concern with rituximab
Targeted synthetic DMARDs	JAK inhibitors: tofacitinib, baricitinib, upadacitinib	Oral options for selected patients after DMARD failure	Serious infection, zoster, thrombosis, cardiovascular and malignancy warnings; CBC/CMP/lipids
Glucocorticoids	Prednisone, methylprednisolone, intra-articular steroid	Short bridge or flare control; intra-articular injection for limited synovitis	Use lowest dose/shortest duration; osteoporosis, infection, diabetes, hypertension, adrenal suppression
NSAIDs/analgesics	Naproxen, ibuprofen, celecoxib, acetaminophen for pain	Symptom relief only	GI bleed, kidney injury, BP/CV risk; do not prevent erosions

RA management algorithm

1. Confirm objective inflammatory synovitis and exclude emergencies: septic arthritis, crystal arthritis, fracture, malignancy, and rapidly progressive systemic disease.
2. Assess baseline disease and drug risk: tender/swollen joint counts, function, ESR/CRP, CBC/CMP, hepatitis/TB risk, pregnancy plans, vaccination status, lung symptoms, kidney/liver disease, and cardiovascular risk.
3. Start a csDMARD early. Methotrexate is favored for many patients with moderate-to-high disease activity when not contraindicated. Give folic acid and counsel on pregnancy avoidance, alcohol moderation, infection precautions, and lab monitoring.
4. Use NSAIDs or a brief glucocorticoid bridge only as needed while DMARDs take effect. Avoid making prednisone the chronic plan.
5. Reassess regularly. If remission or low disease activity is not reached, optimize dose and route, check adherence, then add or switch DMARD therapy.
6. Before biologic or targeted synthetic therapy, screen for latent TB and hepatitis B as appropriate; update vaccines; avoid live vaccines during significant immunosuppression unless planned carefully before therapy.
7. Continue nonpharmacologic care: smoking cessation, exercise/hand therapy, cardiovascular prevention, osteoporosis prevention when steroids are used, and shared decision-making.

Special RA red flags and complications

Presentation	Why it matters	Immediate action
Hot swollen single joint in RA patient	RA and immunosuppression increase septic arthritis risk; flare and gout can look similar	Arthrocentesis before antibiotics if feasible; Gram stain/culture/crystals; urgent antibiotics/drainage if septic suspected
Neck pain, occipital headache, myelopathic signs, planned intubation	Possible atlantoaxial instability	Cervical spine evaluation; avoid forceful neck manipulation; anesthesia alert
Painful red eye or vision change	Scleritis, uveitis, or peripheral ulcerative keratitis can threaten vision	Same-day ophthalmology evaluation
Dyspnea, cough, hypoxemia	Infection, ILD, drug pneumonitis, pleuritis, PE, heart failure	Chest imaging and infection/drug/ILD evaluation; do not assume arthritis flare
Digital ischemia, ulcers, mononeuritis multiplex	Possible rheumatoid vasculitis or antiphospholipid/embolic disease	Urgent rheumatology/vascular/neuro workup; assess systemic vasculitis
Fever on biologic/JAK/steroids	Serious infection may present subtly	Hold immunosuppression depending on severity; evaluate cultures/imaging; treat infection

10B.5 Spondyloarthritis Family: Shared Logic

Spondyloarthritis refers to a family of inflammatory rheumatic diseases linked by axial inflammation, peripheral arthritis, enthesitis, dactylitis, uveitis, psoriasis, inflammatory bowel disease, HLA-B27 association, and a tendency to be seronegative for RF and anti-CCP. Major categories include axial spondyloarthritis, ankylosing spondylitis, psoriatic arthritis, reactive arthritis, enteropathic arthritis, and undifferentiated peripheral spondyloarthritis.

Enthesitis is inflammation where tendons, ligaments, or fascia attach to bone. It can cause heel pain at the Achilles insertion or plantar fascia, lateral epicondyle pain, costochondral pain, pelvic pain, or other focal attachment-site pain. Dactylitis, often called sausage digit, is diffuse swelling of an entire finger or toe due to synovitis, tenosynovitis, and soft-tissue inflammation. These findings point away from classic RA and toward spondyloarthritis, particularly psoriatic arthritis.

SpA feature	Diagnostic meaning	Ask/examine
Inflammatory back pain	Suggests axial SpA when onset <45 and chronic	Improves with exercise, worse with rest, night pain, morning stiffness, alternating buttock pain
Sacroiliitis	Core axial lesion	Buttock pain; pelvic radiograph; MRI SI joints when early disease suspected
Enthesitis	Attachment-site inflammation; common in PsA/SpA	Achilles, plantar fascia, patellar tendon, lateral epicondyle, costochondral sites
Dactylitis	Strong PsA/SpA clue	Whole digit swelling, often toe; ask about prior episodes
Uveitis	Common extra-articular SpA feature	Acute painful red photophobic eye; urgent ophthalmology
Psoriasis/nail disease	Points to PsA	Scalp, behind ears, umbilicus, gluteal cleft; nail pitting/onycholysis
IBD symptoms	Enteropathic arthritis or associated SpA	Chronic diarrhea, blood, abdominal pain, weight loss, perianal disease
Preceding infection	Reactive arthritis	Chlamydia/urethritis/cervicitis; Salmonella, Shigella, Campylobacter, Yersinia, C. difficile

10B.6 Axial Spondyloarthritis and Ankylosing Spondylitis

Axial spondyloarthritis predominantly affects the sacroiliac joints and spine. Ankylosing spondylitis is often used for radiographic axial disease, whereas non-radiographic axial spondyloarthritis has typical clinical and MRI or laboratory features without definite radiographic sacroiliitis. Symptoms usually begin before age 45 and are insidious. The classic back pain improves with exercise, worsens with rest, wakes the patient in the second half of the night, and is accompanied by prolonged morning stiffness. Alternating buttock pain reflects sacroiliac inflammation.

Radiographs may show sacroiliitis, squaring of vertebral bodies, syndesmophytes, and eventual bamboo spine. MRI can detect active sacroiliac inflammation before radiographs become positive. HLA-B27 supports the diagnosis when the clinical pattern is convincing, but it should not be used as a screening test for nonspecific back pain because many HLA-B27 positive individuals never develop SpA. CRP may be elevated but can be normal despite active symptoms.

Axial SpA clue	Why it matters	Diagnostic caveat
Back pain onset before age 45	Degenerative back pain becomes more common with age; young chronic inflammatory back pain raises SpA suspicion	Age cutoff is not absolute; older-onset symptoms require broader differential
Improves with exercise, not rest	Distinguishes inflammatory from mechanical pain	Some mechanical pain also improves with movement; use full pattern
Night pain and morning stiffness	Active inflammation is often worst after inactivity	Sleep disturbance from any pain can mimic night pain
Alternating buttock pain	Suggests sacroiliac involvement	Hip disease, piriformis syndrome, and lumbar radiculopathy can mimic
Acute anterior uveitis	Strong SpA association	Painful photophobic red eye is urgent
Good NSAID response	Many axial SpA patients have rapid improvement	Response is supportive, not diagnostic

Axial SpA treatment

Treatment emphasizes exercise and mobility preservation as much as medication. Physical therapy, daily stretching, posture work, smoking cessation, and avoidance of high-risk spinal manipulation are important. NSAIDs are first-line pharmacologic therapy for pain and stiffness when not contraindicated. If active axial disease persists despite NSAID therapy and objective inflammation is present, biologic therapy such as a TNF inhibitor or IL-17 inhibitor is commonly used. Conventional synthetic DMARDs such as methotrexate and sulfasalazine do not reliably treat axial spine disease, although sulfasalazine may help some peripheral arthritis.

10B.7 Psoriatic Arthritis

Psoriatic arthritis is a heterogeneous spondyloarthritis associated with psoriasis. It can present before, after, or at the same time as skin disease, so absence of known psoriasis does not exclude it. Psoriasis may be hidden on the scalp, behind the ears, in the umbilicus, in the gluteal cleft, or under nails. A family history of psoriasis can be diagnostically useful.

The arthritis pattern may be asymmetric oligoarthritis, symmetric polyarthritis resembling RA, predominant DIP arthritis, axial disease, enthesitis, dactylitis, or arthritis mutilans. Nail disease is a major clue. Pitting, onycholysis, oil-drop discoloration, and nail dystrophy strengthen the diagnosis, particularly when DIP joints are inflamed. Radiographs may show erosions plus new bone formation, pencil-in-cup changes, periostitis, ankylosis, or arthritis mutilans in severe disease.

PsA domain	Clinical findings	Treatment implications
Peripheral arthritis	Oligoarthritis or polyarthritis; may be asymmetric or RA-like	NSAIDs for mild symptoms; csDMARDs such as methotrexate for peripheral disease; biologic/targeted therapy for active or erosive disease
Skin psoriasis	Plaques on extensor surfaces, scalp, umbilicus, intergluteal cleft; severity varies	Coordinate with dermatology; IL-17/IL-23/TNF pathways may treat both skin and joints
Nail disease	Pitting, onycholysis, dystrophy; often linked with DIP disease	Supports diagnosis; may influence biologic choice when severe
Enthesitis	Heel pain, plantar fascia, patellar tendon, epicondyles, chest wall pain	Often responds better to biologic/targeted therapy than to methotrexate alone
Dactylitis	Whole digit swelling; sausage finger/toe	Marker of active PsA and damage risk; evaluate for tenosynovitis and erosions
Axial disease	Inflammatory back pain, sacroiliitis, syndesmophytes	Treat like axial SpA; methotrexate is not reliable for axial disease
IBD/uveitis comorbidity	Crohn/UC symptoms or acute anterior uveitis	Drug selection must consider gut and eye activity; some agents help one domain more than another

PsA vs RA vs OA

Feature	RA	Psoriatic arthritis	Osteoarthritis
Usual pattern	Symmetric MCP/PIP/wrist/MTP polyarthritis	Variable: asymmetric oligoarthritis, DIP disease, RA-like polyarthritis, axial disease, dactylitis	DIP, PIP, first CMC, knees, hips; mechanical pattern
DIP involvement	Uncommon/prominent DIP argues against typical RA	Common, especially with nail disease	Common with Heberden nodes
Morning stiffness	Often >30-60 min	Often inflammatory if active	Usually <30 min; gelling after inactivity possible
Skin/nails	Nodules in seropositive disease	Psoriasis, nail pitting, onycholysis	No inflammatory skin/nail pattern
Serology	RF and/or anti-CCP often positive	Usually RF/anti-CCP negative; anti-CCP rarely positive	Negative inflammatory serology
Radiographs	Marginal erosions, periarticular osteopenia, symmetric joint-space loss	Erosions plus new bone formation, pencil-in-cup, ankylosis, periostitis	Osteophytes, subchondral sclerosis/cysts, asymmetric cartilage loss

PsA treatment overview

Psoriatic arthritis treatment is domain-driven. Mild peripheral symptoms may respond to NSAIDs or local glucocorticoid injection, but persistent inflammatory arthritis should not be treated indefinitely with analgesics alone. Methotrexate is often used for peripheral arthritis and skin disease, though biologic or targeted synthetic therapy may be preferred for active enthesitis, dactylitis, axial disease, erosive disease, or inadequate response. TNF inhibitors, IL-17 inhibitors, IL-12/23 or IL-23 pathway agents, PDE-4 inhibition, and JAK inhibition are selected based on arthritis pattern, skin burden, IBD, uveitis, infection risk, pregnancy plans, and prior drug response.

10B.8 Reactive Arthritis

Reactive arthritis is an inflammatory arthritis triggered by infection elsewhere, classically a genitourinary infection such as Chlamydia trachomatis or an enteric infection such as Salmonella, Shigella, Campylobacter, Yersinia, or Clostridioides difficile. The arthritis is sterile in the joint, meaning viable organisms are usually not cultured from synovial fluid, although immune activation follows infection. Symptoms often begin days to weeks after infection. The classic triad of arthritis, urethritis, and conjunctivitis is memorable but insensitive; many patients lack one or more elements.

The typical arthritis is asymmetric oligoarthritis affecting large lower-extremity joints and toes. Enthesitis, Achilles tendon pain, plantar fasciitis, dactylitis, low back pain, sacroiliitis, conjunctivitis, anterior uveitis, oral ulcers, circinate balanitis, cervicitis or urethritis, and keratoderma blennorrhagicum can occur. HLA-B27 is associated with more severe or chronic disease but is not required for diagnosis.

Reactive arthritis feature	Clinical details	Management point
Trigger	Chlamydia urethritis/cervicitis; enteric infections including Salmonella, Shigella, Campylobacter, Yersinia, C. difficile	Test for active STI when suspected; treat patient and partners according to STI guidance
Joint pattern	Asymmetric oligoarthritis, usually lower extremity; knee, ankle, toes; dactylitis	Arthrocentesis if effusion raises concern for septic/crystal arthritis

Reactive arthritis feature	Clinical details	Management point
Eye disease	Conjunctivitis or anterior uveitis	Painful photophobic red eye requires urgent ophthalmology
Mucocutaneous disease	Oral ulcers, circinate balanitis, keratoderma blennorrhagicum	Look for lesions; may resemble psoriasis
Enthesitis/axial symptoms	Achilles, plantar fascia, inflammatory back pain, sacroiliitis	Treat like peripheral/axial SpA depending on dominant domain
Therapy	NSAIDs first; local steroids; DMARDs such as sulfasalazine or methotrexate for chronic disease; biologic therapy in refractory cases	Antibiotics treat active infection but may not rapidly resolve established arthritis

10B.9 Diagnostic Algorithms

Algorithm 1: symmetric inflammatory polyarthritis

1. Confirm objective synovitis in at least one joint. Without swelling, warmth, effusion, or synovitis-limited motion, consider arthralgia syndromes and close follow-up.
2. Ask duration. Symptoms shorter than six weeks can be viral, reactive, crystal, early RA, or early connective tissue disease. Symptoms longer than six weeks with MCP/PIP/wrist/MTP synovitis strongly raise RA probability.
3. Order RF, anti-CCP, ESR/CRP, CBC, CMP, and baseline radiographs or ultrasound when RA is suspected. Do not order broad autoantibody panels without clinical clues.
4. Look for mimics: psoriasis/nails/dactylitis/enthesitis; lupus features; viral exposure; hepatitis risk; crystal disease; hemochromatosis; sarcoid; paraneoplastic features.
5. Refer early to rheumatology and start disease-modifying therapy when RA is likely or confirmed.

Algorithm 2: inflammatory back pain in a young adult

1. Classify the pain pattern: onset before 45, insidious onset, morning stiffness, improvement with exercise, worsening with rest, nocturnal pain, alternating buttock pain.
2. Search for SpA features: uveitis, psoriasis, nail disease, IBD, enthesitis, dactylitis, family history, good NSAID response, prior reactive arthritis.
3. Obtain pelvic radiograph for sacroiliitis if symptoms are chronic and suspicion is moderate/high. If radiographs are negative but suspicion remains high, MRI of sacroiliac joints can detect active inflammation.
4. Use HLA-B27 selectively. A positive test supports SpA in the right context, but a positive test alone does not diagnose disease and a negative test does not exclude disease.
5. Begin exercise/physical therapy and NSAIDs when safe. Escalate to biologic or targeted therapy for active disease with objective inflammation despite first-line treatment.

Algorithm 3: suspected psoriatic arthritis

1. Ask every inflammatory arthritis patient about psoriasis, family history of psoriasis, nail changes, heel pain, dactylitis, inflammatory back pain, uveitis, and IBD symptoms.
2. Perform a deliberate skin and nail exam: scalp, behind ears, elbows/knees, umbilicus, intergluteal cleft, groin folds, nails.
3. Determine active domains: peripheral arthritis, axial disease, enthesitis, dactylitis, skin, nail, IBD, uveitis.
4. Use RF and anti-CCP to assess RA mimic but do not exclude PsA solely because RF is weakly positive.
5. Choose therapy by domain and severity. Methotrexate may fit peripheral arthritis/skin disease; biologic or targeted therapy is often needed for erosive, axial, entheseal, or severe skin disease.

Algorithm 4: post-infectious oligoarthritis

1. First exclude emergencies: septic arthritis, disseminated gonococcal infection, crystal arthritis, fracture, and cellulitis-adjacent joint infection.
2. Ask about diarrhea or genitourinary symptoms one to six weeks before arthritis. Ask about sexual exposures, dysuria, discharge, cervicitis symptoms, abdominal pain, fever, and antibiotics.
3. Look for conjunctivitis, uveitis, oral ulcers, keratoderma blennorrhagicum, circinate balanitis, enthesitis, and dactylitis.

4. Test for Chlamydia/gonorrhea by NAAT when STI trigger is possible; stool studies are most useful if GI infection is ongoing or public health implications exist.
5. Treat active infection appropriately. Use NSAIDs for arthritis if safe, consider local steroids for selected joints, and escalate to DMARD/biologic therapy only if chronic or severe inflammatory disease persists.

10B.10 Disease-by-Disease Treatment and Monitoring Matrix

Disease	First-line concepts	Escalation concepts	Monitoring and red flags
RA	Early csDMARD, often methotrexate when appropriate; short steroid bridge only if necessary; NSAIDs for symptoms	Add/switch csDMARDs, biologic DMARDs, or JAK inhibitors using treat-to-target strategy	CBC/CMP, ESR/CRP, disease activity, infection screening, vaccine status, lung disease, cervical spine, eye symptoms, cardiovascular risk
Axial SpA/AS	Exercise/PT, posture, smoking cessation, NSAIDs if safe	TNF inhibitor, IL-17 inhibitor, or other targeted therapy for active objective inflammation despite NSAIDs	Uveitis, IBD, psoriasis, spinal fracture risk in fused spine, aortic regurgitation, restrictive chest wall disease
PsA	NSAIDs/local injections for mild disease; methotrexate or other csDMARD for peripheral arthritis in selected patients	Domain-based biologic/targeted therapy: TNF, IL-17, IL-23/IL-12/23, JAK/PDE4 depending on skin/joint/IBD/uveitis profile	Skin/nail burden, enthesitis/dactylitis, erosions, IBD/uveitis, metabolic syndrome, infection risk
Reactive arthritis	Treat active trigger infection; NSAIDs; local steroids when appropriate; PT	Sulfasalazine/methotrexate for chronic peripheral arthritis; biologic therapy for refractory severe disease	Rule out septic joint; STI partner treatment; uveitis; chronic axial/peripheral disease

10B.11 Immunosuppression Safety

Many therapies used in RA and spondyloarthritis alter host defense. Before escalating to biologic or targeted synthetic therapy, clinicians typically review latent tuberculosis risk, hepatitis B status when relevant, hepatitis C/HIV risk when relevant, vaccination status, history of recurrent infections, prior malignancy, demyelinating disease, heart failure, pregnancy plans, and baseline cytopenias or liver disease. Fever or focal infection in an immunosuppressed patient may be muted, and inflammatory markers can be affected by the drug mechanism. For example, IL-6 blockade can blunt CRP responses, so clinical assessment remains essential.

Safety issue	Why it matters	Practical approach
Vaccines	Immunosuppressed patients have higher infection risk and may have reduced vaccine response	Update non-live vaccines before therapy when possible; avoid live vaccines during significant immunosuppression unless planned before treatment
Latent TB	TNF inhibitors and other immunosuppressants can reactivate TB	Screen based on planned therapy and risk; treat latent TB before high-risk immunosuppression
Hepatitis B	Rituximab and other immunosuppression can reactivate HBV	Check HBV serologies when indicated; antiviral prophylaxis/monitoring for at-risk patients
Baseline labs	DMARDs can cause cytopenia, hepatotoxicity, renal dosing issues	CBC, CMP, kidney/liver assessment before and during therapy
Pregnancy	Methotrexate and leflunomide are teratogenic; drug selection differs in pregnancy/lactation	Ask before prescribing; use pregnancy-compatible regimens when needed
Steroid toxicity	Infection, osteoporosis, hyperglycemia, hypertension, mood changes, cataracts, avascular necrosis, adrenal suppression	Use lowest dose/shortest time; bone protection when prolonged; taper thoughtfully

10B.12 High-Yield Rapid Review

Question stem clue	Most likely diagnosis	Why
Middle-aged woman with symmetric MCP/PIP/wrist swelling, morning stiffness >1 hour, anti-CCP positive	Rheumatoid arthritis	Classic symmetric small-joint inflammatory synovitis with specific serology
Young man with chronic back pain improved by exercise, alternating buttock pain, uveitis, HLA-B27 positive	Axial spondyloarthritis	Inflammatory back pain plus extra-articular SpA clue
Patient with DIP arthritis, nail pitting, dactylitis, and plaques in scalp	Psoriatic arthritis	DIP/nail/dactylitis pattern points away from RA and toward PsA
Asymmetric knee/ankle arthritis after urethritis, conjunctivitis, heel pain	Reactive arthritis	Post-GU infection with lower-extremity oligoarthritis and enthesitis
Long-standing seropositive RA with neutropenia and splenomegaly	Felty syndrome	Classic extra-articular hematologic RA complication
RA patient with painful red eye	Scleritis until proven otherwise	Vision-threatening extra-articular inflammatory disease
Fused spine patient after minor trauma with new neurologic symptoms	Spinal fracture/instability in ankylosing spondylitis	Rigid ankylosed spine fractures easily and can injure cord

Question stem clue	Most likely diagnosis	Why
RA patient on biologic with fever and hot knee	Septic arthritis until proven otherwise	Immunosuppression changes risk; flare is diagnosis of exclusion

Disease	Serology/lab pattern	Imaging pattern	One-line discriminator
RA	RF and/or anti-CCP often positive; ESR/CRP often elevated	Marginal erosions, periarticular osteopenia, symmetric joint-space loss	Symmetric small-joint synovitis with DIP sparing
Axial SpA/AS	HLA-B27 often positive; CRP may be elevated or normal; RF/anti-CCP negative	Sacroiliitis, syndesmophytes, bamboo spine; MRI marrow edema early	Inflammatory back pain before 45 with sacroiliac/extra-articular features
PsA	Usually RF/anti-CCP negative; inflammatory markers variable	Erosions plus new bone formation, pencil-in-cup, DIP disease, ankylosis	Psoriasis/nails/dactylitis/enthesitis with variable arthritis
Reactive arthritis	No diagnostic antibody; HLA-B27 may predict severity; cultures/NAAT depend on trigger	May show enthesitis, periostitis, sacroiliitis in chronic disease	Sterile arthritis after GU/GI infection

10B.13 Integrated Mini-Cases

Case 1: early RA

A 42-year-old woman has eight weeks of bilateral wrist, MCP, and PIP swelling with morning stiffness lasting two hours. RF is weakly positive and anti-CCP is strongly positive. ESR and CRP are elevated. Hand radiographs are normal. Normal radiographs do not exclude RA because erosions may not yet be visible. This patient needs early DMARD therapy and monitoring rather than reassurance.

Case 2: RA mimic by PsA

A 50-year-old man has hand pain and swelling. He has DIP tenderness, nail pitting, heel pain, and a history of scaly scalp plaques. RF is negative. The pattern is more consistent with psoriatic arthritis than RA. A focused skin and nail exam prevents misclassification.

Case 3: axial SpA

A 28-year-old man has three years of low back pain that wakes him at night and improves after exercise. He reports a prior painful red eye. Pelvic radiograph is unrevealing, but MRI shows active sacroiliitis. This is compatible with non-radiographic axial spondyloarthritis; absence of radiographic bamboo spine does not exclude early disease.

Case 4: reactive arthritis

A 24-year-old develops asymmetric ankle and knee swelling three weeks after urethral discharge. He also has conjunctivitis and Achilles insertion pain. Test for Chlamydia and gonorrhea, treat active infection and partners as appropriate, and use NSAIDs or local steroid therapy for arthritis after septic arthritis is excluded.

Case 5: dangerous RA complication

A patient with long-standing erosive RA presents for surgery and reports neck pain and intermittent hand numbness. Atlantoaxial instability must be considered before intubation. This is a procedural safety issue, not a routine joint-pain complaint.

10B.14 Graph-RAG Concept Map Seeds

For graph extraction, important concept nodes include rheumatoid arthritis, synovitis, pannus, anti-CCP, rheumatoid factor, marginal erosion, morning stiffness, MCP arthritis, PIP arthritis, DIP sparing, rheumatoid nodule, Felty syndrome, atlantoaxial subluxation, rheumatoid lung disease, scleritis, treat-to-target, methotrexate, biologic DMARD, JAK inhibitor, axial spondyloarthritis, ankylosing spondylitis, inflammatory back pain, sacroiliitis, syndesmophyte, HLA-B27, enthesitis, dactylitis, psoriatic arthritis, nail pitting, psoriasis, reactive arthritis, Chlamydia trigger, enteric trigger, anterior uveitis, and keratoderma blennorrhagicum.

Edge	Relationship type	Rationale
Anti-CCP -> Rheumatoid arthritis	diagnostic marker / prognostic marker	Anti-CCP is highly specific and predicts erosive RA in the correct clinical context
Synovitis -> Pannus -> Erosion	pathophysiology sequence	Chronic synovial inflammation creates invasive tissue that damages cartilage and bone
RA -> Atlantoaxial instability	complication	Cervical synovitis and ligament injury can destabilize C1-C2
Psoriasis/nail pitting -> Psoriatic arthritis	clinical association	Skin/nail disease plus inflammatory arthritis suggests PsA
Enthesitis -> Spondyloarthritis	clinical pattern	Attachment-site inflammation is a core SpA feature
Inflammatory back pain -> Axial spondyloarthritis	diagnostic clue	Young-onset back pain improved by exercise and worse with rest suggests axial inflammation
Chlamydia infection -> Reactive arthritis	trigger relationship	GU infection can precede sterile inflammatory arthritis

Edge	Relationship type	Rationale
Biologic DMARD -> Infection risk	treatment adverse effect	Targeted immune suppression increases infection/reactivation risk

Selected References and Source Anchors

Uploaded source: rheum.pdf, pages 16-20, especially RA general characteristics, clinical features, cervical instability, rheumatoid nodules, RF/anti-CCP discussion, and extra-articular RA manifestations; pages 15-16 frame overlap disease and RA differential clues.

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